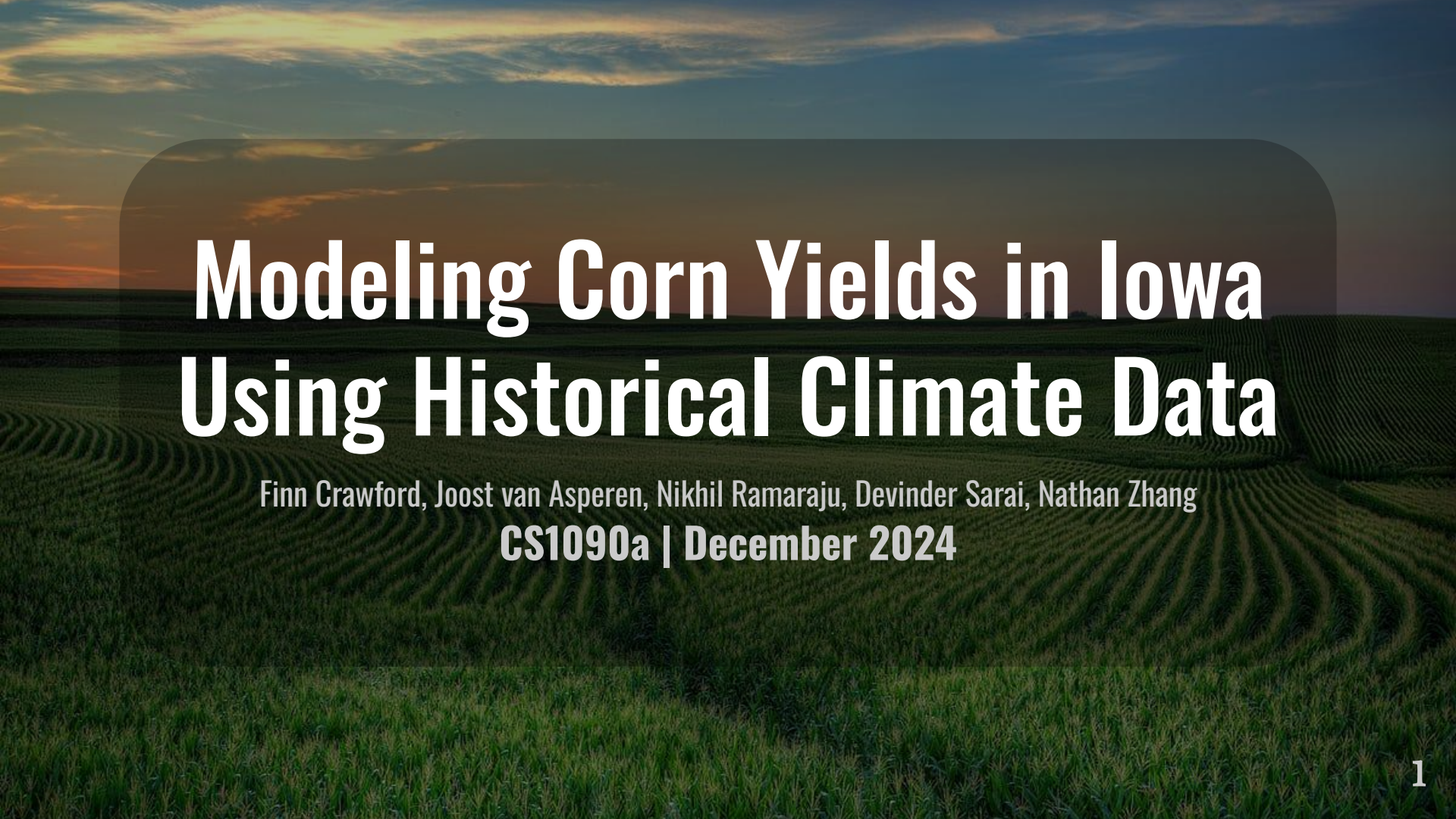




Modeling Corn Yields in Iowa Using Historical Climate Data

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Research Question:

**Can climate data be used to
predict corn yields in Iowa?**



Motivation

- **Climate** affecting regions in unpredictable ways.
- Disrupting the conditions needed for efficient large-scale crop production.
- Models to understand and quantify the relationship between climate and **crop yield**.
- Growing human population requires increased agricultural yields.
- Can agricultural output keep up with growing demand despite climate change?

Data Sources

Climate Data (NOAA)



- Daily climate measurements 1926–2022
- Includes maximum/minimum temperature (°C) and precipitation (in)
- Ten stations across Iowa
- n = 356598

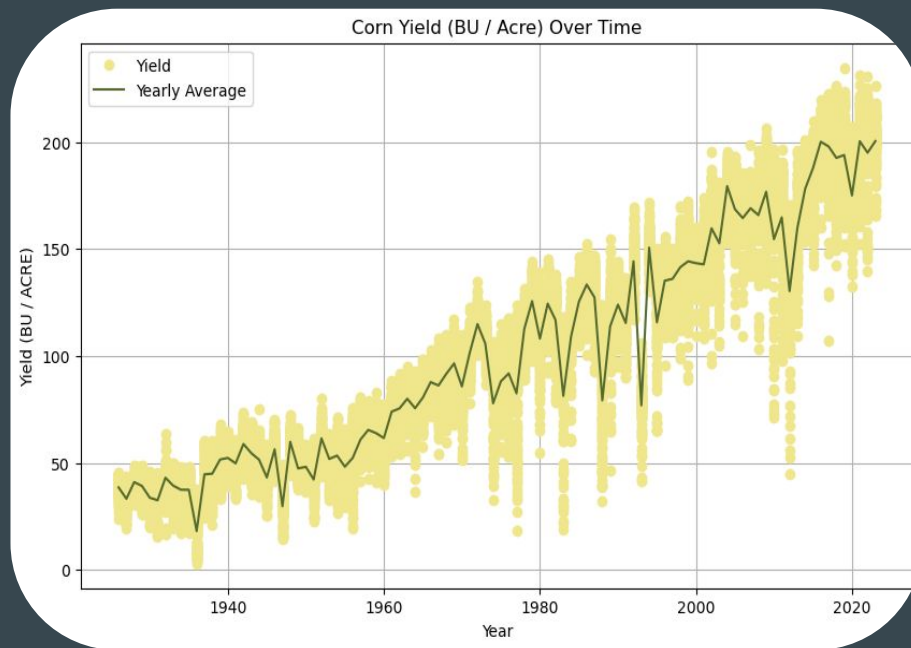
Agricultural Data (USDA)



- Annual county-level corn yields 1926–2022
- Measured in Bushels/Acre
- 98 counties across Iowa
- n = 9658

Exploratory Data Analysis

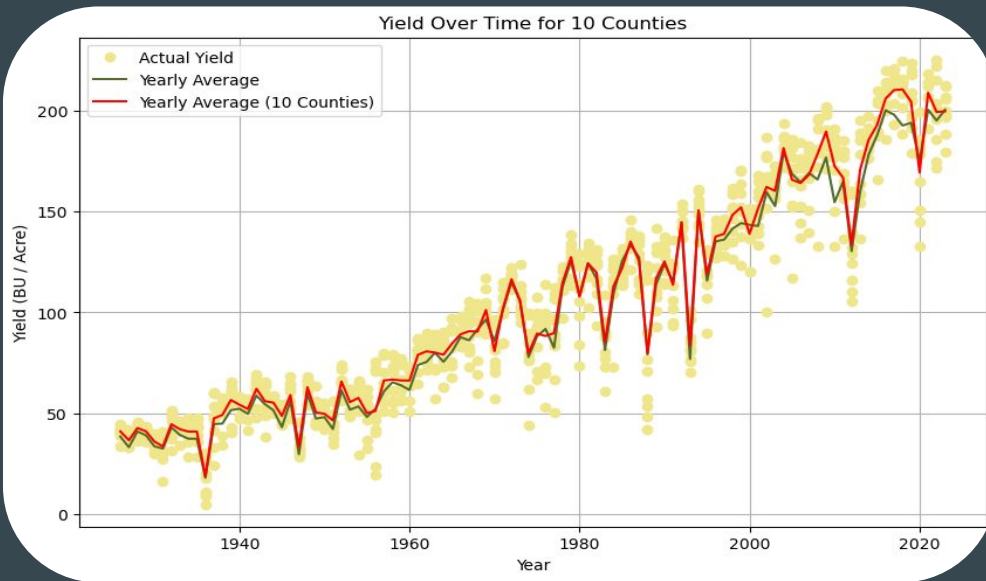
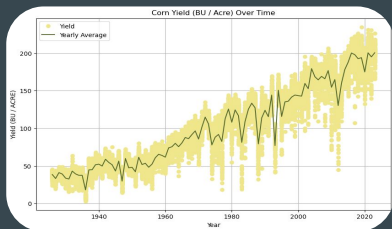
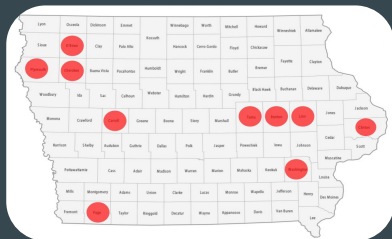
- Clear trend of improving yield over time (~30 → 170 Bu/Acre)
- Increase likely due to improvements in agricultural technology (Dimitri et al., 2005)
- Using 'year' as an additional feature to proxy for improvements in technology



Data Preprocessing



Weather Station Coverage



- Cut down crop yield data to just 10 the counties containing weather stations
- Observed minimal change in yearly average trend between original and cropped data
- Encode the 10 selected countries into binary columns using **OneHotEncoding**

Climate Parameters

1. Constrain data to Growing Season (May 1 – October 15th)

PRCP

TMAX

TMIN

2. Create custom parameters using existing data:

GDD

“Growing Degree Days”

$$(T_{\text{Max}} + T_{\text{min}})/2 - 50^{\circ}\text{F}$$

A measure of heat accumulation used to estimate plant growth

Heat Shock

of days with:
 $T_{\text{max}} > 90^{\circ}\text{F}$

Cold Shock

of days with:
 $T_{\text{min}} < 32^{\circ}\text{F}$

PRCP

$\Sigma(\text{PRCP})$ during growing season

3. Create interaction term **GDD x PRCP**

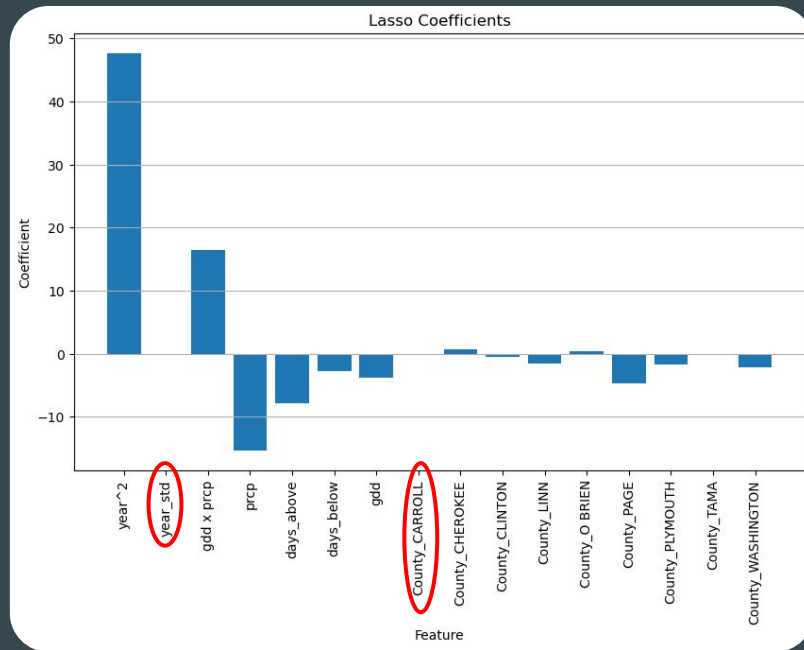
4. Standardize final terms

Baseline Model (LASSO Regression)



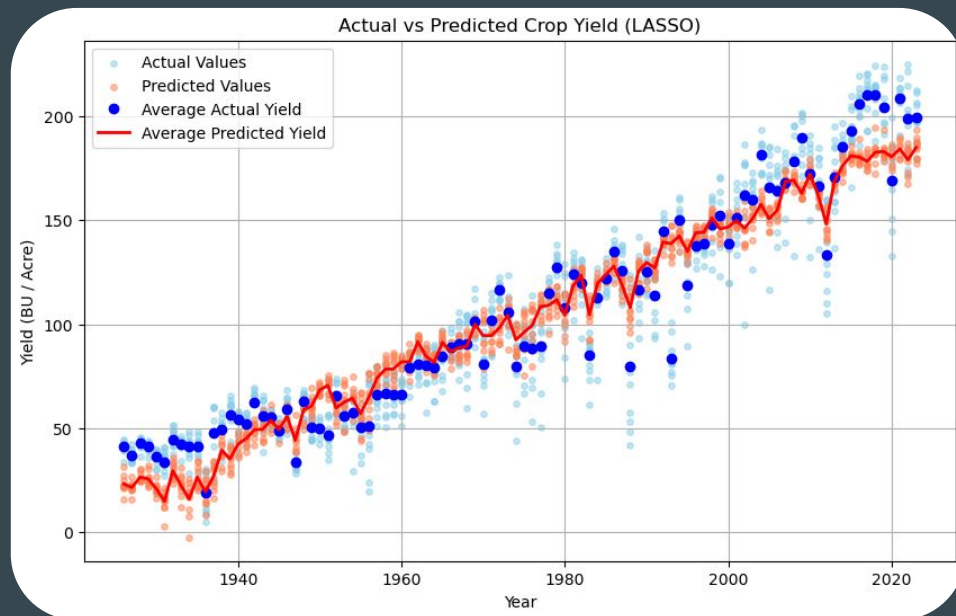
LASSO Coefficients

- Most important predictor is **Year²**
 - Improvements in technology have led to the greatest increase in yield
- **Year** parameter shrunk to 0
 - High degree of multicollinearity with **Year²**
- Environmental predictors such as **gdd x prcp** and **prcp** have high coefficients as well
- **County_CARROLL** also shrunk to 0
 - Likely a High degree of similarity with the reference county (**BENTON**)

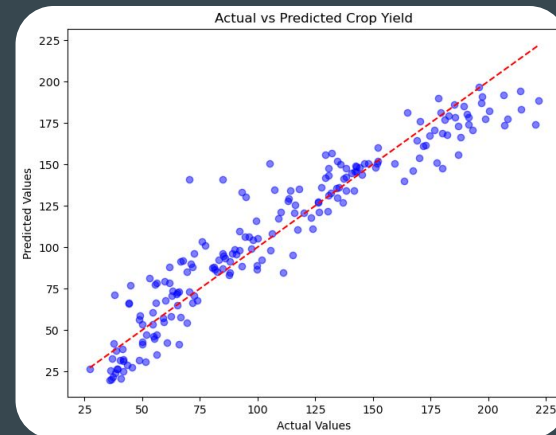


(cv=5, $\alpha=0.05$)

LASSO Results: Prediction

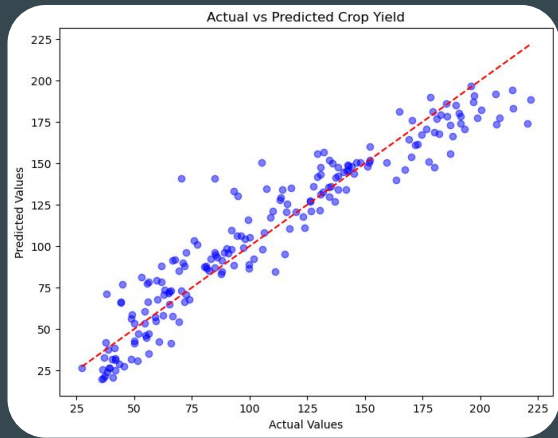
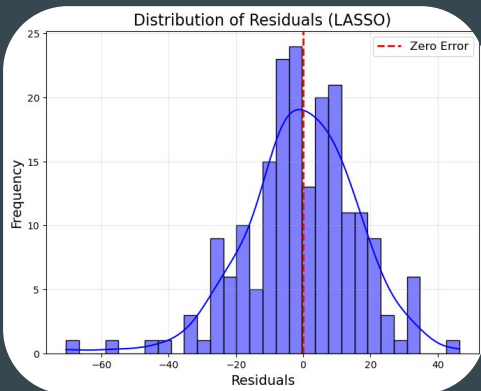
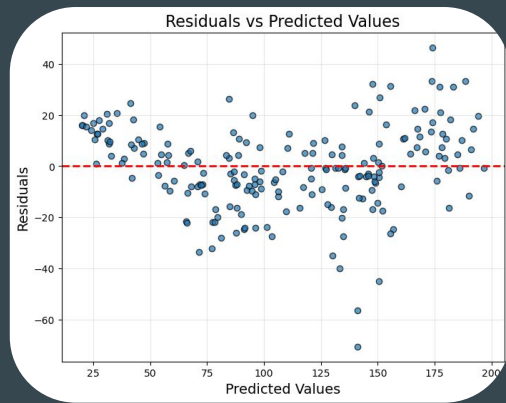


- Model achieves an R^2 value of ~ 0.9 , meaning the model accounts for roughly 90% of observed variance
- However, the predictions appear more stable than reality, and the model seems unable to capture some of the 'dips' in yield



MSE: 272.51
R-squared: 0.898

LASSO Results: Error



- Error appears somewhat **heteroskedastic**, and residual distribution is **left-skewed**, suggesting the model is systematically failing to predict 'dips' in yield at higher values
- In this case, a **Random Forest** model may be a good next step to capture potential non-linear relationships driving this discrepancy

MSE: 272.51
R-squared: 0.898

Final Model

(Random Forest Regression)



Model Pipeline

Hyperparameter Tuning

- Define a **parameter grid** using specified values
- Find optimal hyperparameter combination using **GridSearchCV** five-fold cross-validation

Model Training

- Train a **single decision tree** using optimal parameters
- Train a **random forest regression** model with the same parameters

Model Evaluation

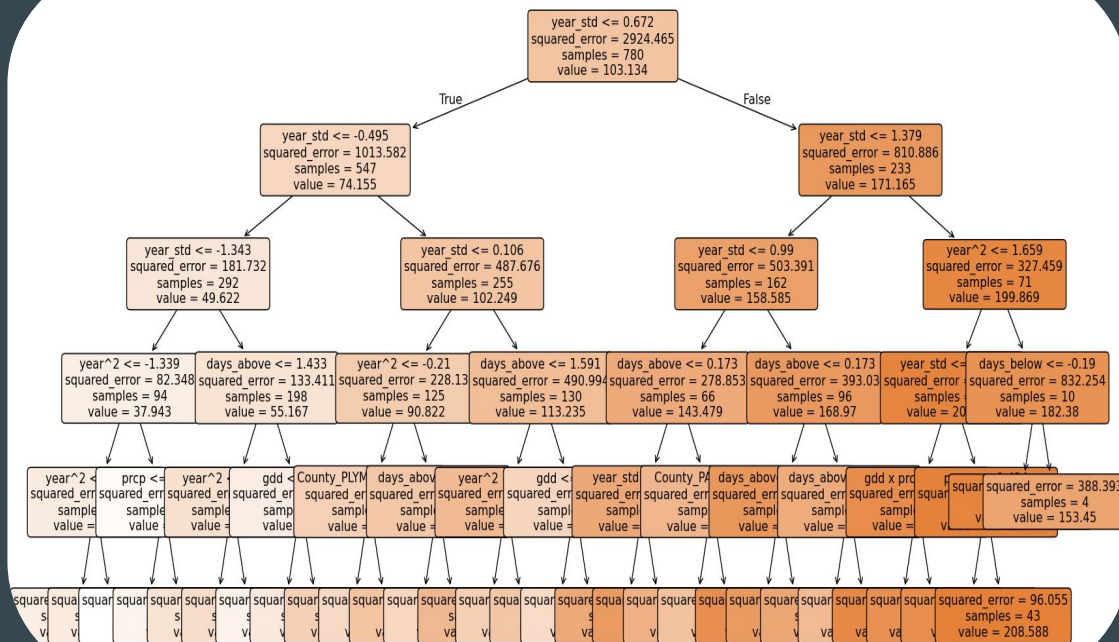
- Calculate **MSE** and **R²** for both models to assess whether the increased complexity of RF is justified
- Use single DT to interpret parameter relationships

Visualization/Analysis

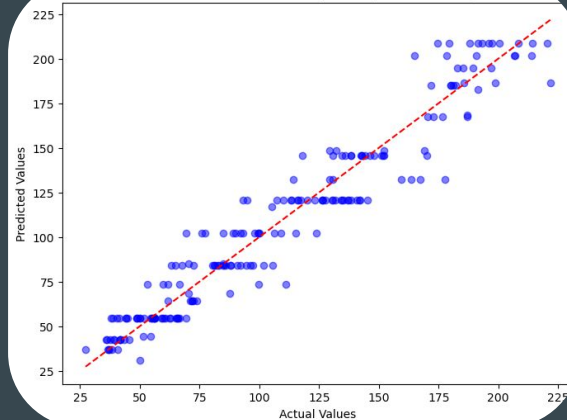
- Visualize RF feature importances
- Plot RF predictions against annual yield trend

Single Decision Tree

Regression Decision Tree

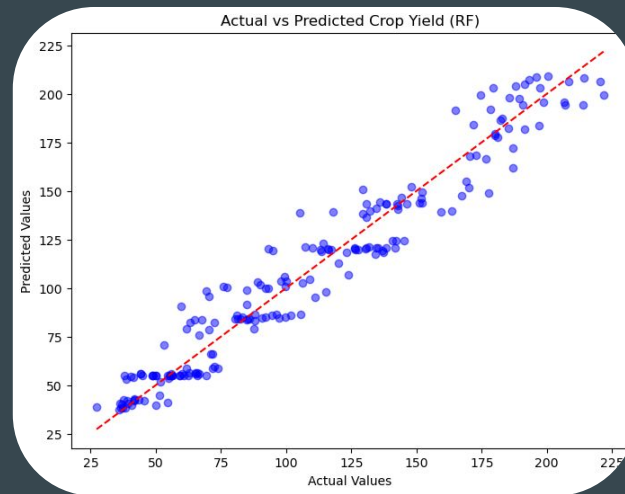
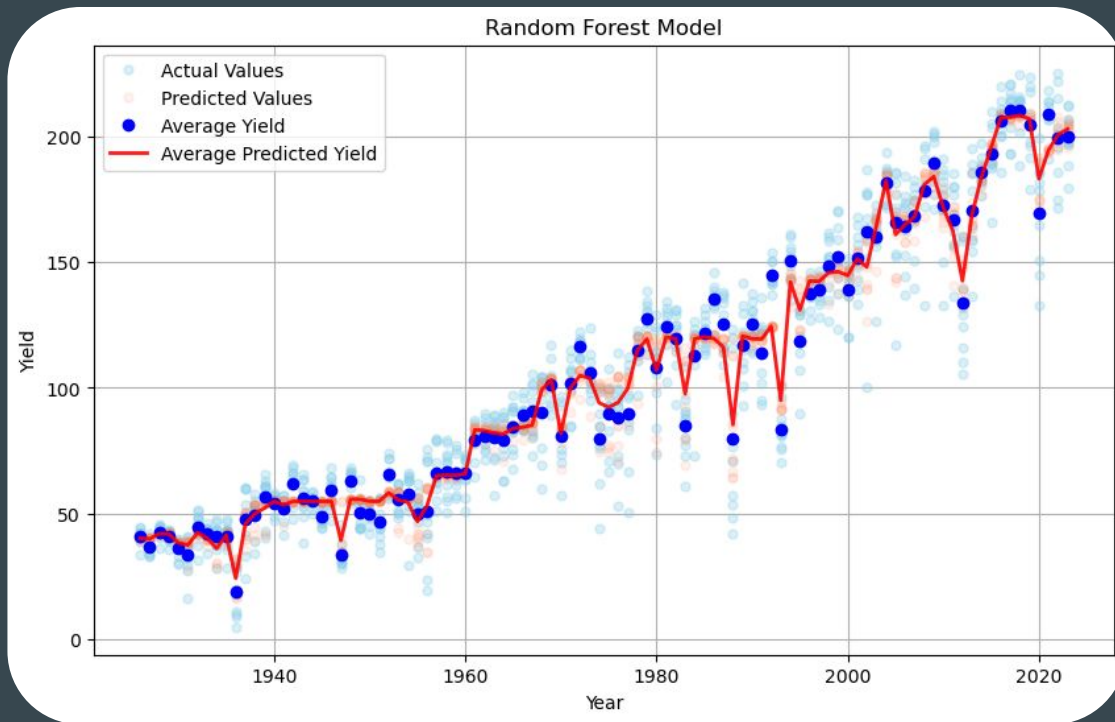


Actual vs Predicted Crop Yield (Decision Tree)



MSE: 190.46
R-squared: 0.929

Random Forest



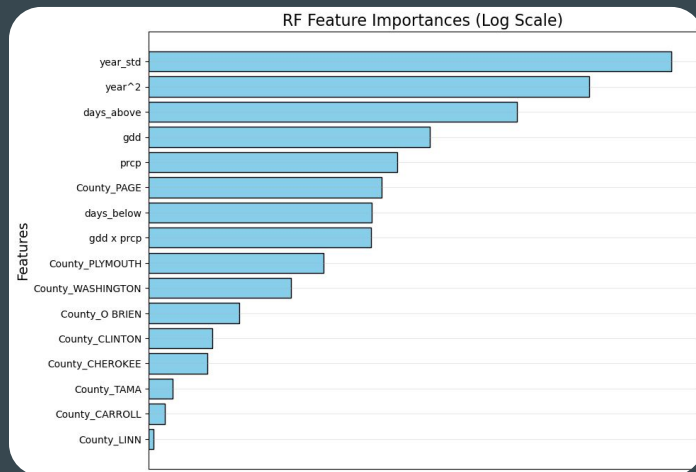
DT
MSE: 190.46
 R^2 : 0.929

RF
MSE: 123.54
 R^2 : 0.954

Feature Importances (RF)

- **Year** and **Year²** are once again the most important features, indicating a strong reliance on technological advancement
- **Days_above** is the next most important feature, demonstrating the potential negative implications of heat shock on corn crops
- **GDD** and **prcp** then follow, showing that precipitation and temperature still play key roles in crop yield

```
year_std: 0.8216918855286927
year^2: 0.13777640983272774
days_above: 0.029065831790568313
gdd: 0.004382772567653218
prcp: 0.0021843109857726368
County_PAGE: 0.0015498456473096531
days_below: 0.0012616578560102189
gdd x prcp: 0.001238265924494535
County_PLYMOUTH: 0.000441317647250491
County_WASHINGTON: 0.00021847054219067489
County_O_BRIEN: 7.13976925926274e-05
County_CLINTON: 4.007092216829735e-05
County_CHEROKEE: 3.560568623646193e-05
County_TAMA: 1.6865623605664803e-05
County_CARROLL: 1.4226613682776709e-05
County_LINN: 1.1065139044097589e-05
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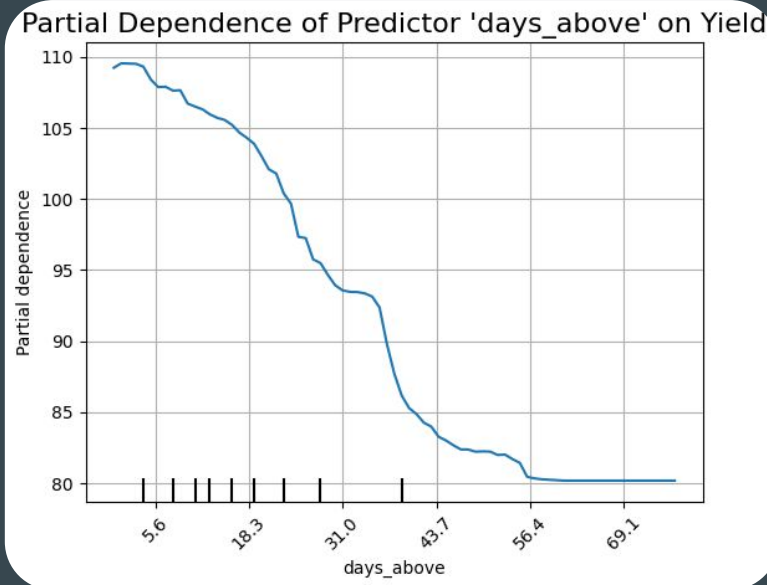


Sensitivity Analysis



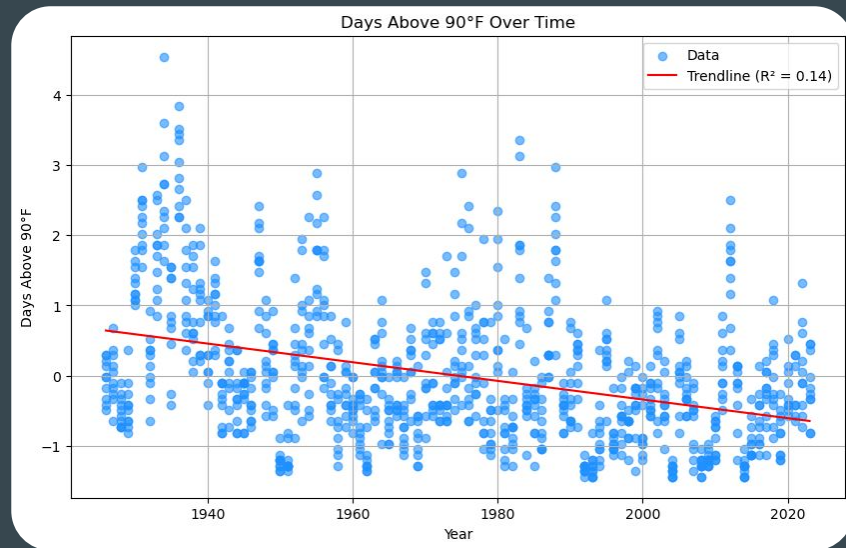
Partial Dependence Analysis

- Given **days_above** was the most important climate parameter in the RF mode, we elected to perform partial dependence analysis to further quantify its importance
- We find that the predicted yield changes greatly as we change the **days_above** variable
- This may be alarming, as climate change will likely increase this predictor



A Word to Stakeholders

- Although our results from the partial dependence analysis may seem frightening, we see from the plot on the right that **days_above** exhibits no strong trends over the past century ($R^2 = 0.14$).
- Our trendline shows a weak decrease in the **days_above** predictor over time, in line with a slight decrease in **GDD** over the same period.
- However, using information from our previous analysis, stakeholders should still be wary about the implications of temperature increase on crop yield.



Sources

1. Carolyn Dimitri, Anne Effland, and Neilson Conklin. The 20th Century Transformation of U.S. Agriculture and Farm Policy. Economic Information Bulletin No. 3, United States Department of Agriculture, Economic Research Service, June 2005.
2. "Factors Impacted by Increasing Plant Populations in Corn." Crop Science US, Bayer, <https://www.cropscience.bayer.us/articles/bayer/factors-impacted-by-increasing-plant-populations-in-corn>. Accessed 2 Dec. 2024.
3. Zachary Curtis, Brittany Clark. Penn State Extension. "Understanding Growing Degree Days." Penn State Extension, [https://extension.psu.edu/understanding-growing-degree-days#:~:text=Growing%20degree%20days%20\(GDD\)%2C,reach%20a%20predetermined%20growth%20stage](https://extension.psu.edu/understanding-growing-degree-days#:~:text=Growing%20degree%20days%20(GDD)%2C,reach%20a%20predetermined%20growth%20stage). Accessed 2 Dec. 2024.
4. Lesk, Corey, et al. "Influence of Extreme Weather Disasters on Global Crop Production." Nature, vol. 529, 2016, pp. 84–87. <https://www.nature.com/articles/nature16467>
5. Deutsch, Curtis A., et al. "Increase in Crop Losses to Insect Pests in a Warming Climate." Science, vol. 361, no. 6405, 2018, pp. 916–919. <https://www.science.org/doi/10.1126/science.aat3466>
6. McKinsey & Company. "Agriculture's Connected Future: How Technology Can Yield New Growth." McKinsey & Company, October 2020. <https://www.mckinsey.com/industries/agriculture/our-insights/agricultures-connected-future-how-technology-can-yield-new-growth>
7. U.S. Department of Agriculture Economic Research Service. "Innovations in Seed and Farming Technologies Drive Productivity Gains and Costs on Corn Farms." USDA ERS, April 2022. <https://www.ers.usda.gov/amber-waves/2022/april/innovations-in-seed-and-farming-technologies-drive-productivity-gains-and-costs-on-corn-farms/>

Thank You

